

VICTORIA JUNIOR COLLEGE
2020 JC1 PROMOTIONAL EXAMS
H2 CHEMISTRY PAPER 1 ANSWERS

1	B	6	B	11	B	16	D	21	D	26	A
2	C	7	B	12	C	17	A	22	A	27	B
3	B	8	C	13	C	18	A	23	B	28	A
4	A	9	D	14	C	19	A	24	D	29	B
5	C	10	A	15	D	20	D	25	C	30	D

1 B

angle of deflection $\propto \frac{\text{charge}}{\text{mass}}$

The $\frac{\text{charge}}{\text{mass}}$ ratio for ${}^9\text{Be}^{2+}$ is $\frac{+2}{9}$

	particle	$\frac{\text{charge}}{\text{mass}}$
A	${}^{14}\text{C}^{4-}$	$\frac{-4}{14} = \frac{-2}{7}$
B	${}^{18}\text{O}^{2-}$	$\frac{-2}{18} = \frac{-1}{9}$
C	${}^{20}\text{Mg}^{2+}$	$\frac{+2}{20} = \frac{+1}{10}$
D	${}^{45}\text{P}^{5+}$	$\frac{+5}{35} = \frac{+1}{7}$

2 C

Since Z^{3+} has 19 electrons, Z has 22 electrons.
 This means that Z is Ti.

${}_{22}\text{Ti}: 1s^2 2s^2 2p^6 3s^2 3p^6 3d^2 4s^2$

Option A (Incorrect)

Element Z is a d-block element in Group 4.

Option B (Incorrect)

Element Z is in Period 4.

Option C (Correct)

Element Z is a d-block element.

Option D (Incorrect)

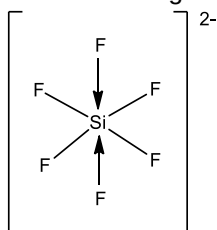
Element Z has 2 half-filled d orbitals.

3 B

Si, Ge and C are all from Group 14.

Option A: True

There are 6 sigma bonds in SiF_6^{2-} .



Option B: False

F^- bonds to SiF_4 via dative covalent bonds.

Option C: True

See structure above for SiF_6^{2-} .

Option D: True

C is in Period 2 and does not have energetically available d orbitals to expand octet to form CF_6^{2-} .

4 A (1, 2 and 3)

ICl_2^- : linear (2 bp, 3 lp), planar, 180°

ClF_3 : T-shaped (3 bp, 2 lp), planar, 90°

ClO_3^- : trigonal pyramidal (3 bp, 1 lp), 107°

Option 1: Correct

The bond angles of $\text{ClF}_3 < \text{ClO}_3^- < \text{ICl}_2^-$.

Option 2: Correct

ClF_3 and ClO_3^- have a net dipole moment. Only

ICl_2^- is non-polar.

Option 3: Correct

Only ClO_3^- is not planar.

5 C (3 and 4 only are correct)

Option 1 (Wrong)

There is no hydrogen bonds between the nonpolar hydrocarbon molecules, only instantaneous dipole-induced dipole interactions.

Option 2 (Wrong)

During boiling, only the instantaneous dipole-induced dipole interactions between the hydrocarbon molecules are broken. The covalent bonds remain intact.

Option 3 (Correct)

There is more surface area for molecular interactions in the straight-chained butane than in the branched 2-methylpropane, resulting in closer packing due to the larger extent and stronger instantaneous dipole-induced dipole interactions. Hence, more energy is required to overcome the stronger intermolecular forces and the boiling point is higher for butane than 2-methylpropane.

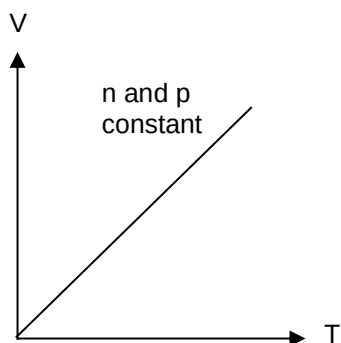
Option 4 (Correct)

As the M_r increases and the number of electrons increases, the strength of the instantaneous dipole-induced dipole interactions increases. The energy required to break them increases resulting in a higher boiling point.

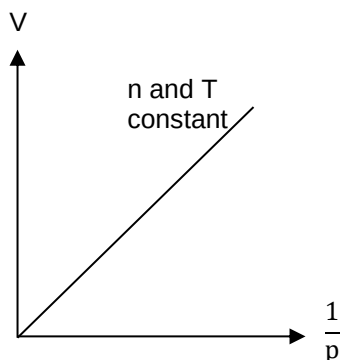
6 B

Cations with higher charge and smaller ionic radius have higher charge density and polarise the anion electron cloud more to form a more covalent bond. For ionic compounds of Group 14 halides, we look for cation with smaller +2 charge and larger ionic radius ($\text{Pb}^{2+} > \text{Sn}^{2+}$ in size), hence Pb^{2+} . Since F^- has smaller electron cloud than Cl^- and are less easily polarised, PbF_2 is most ionic.

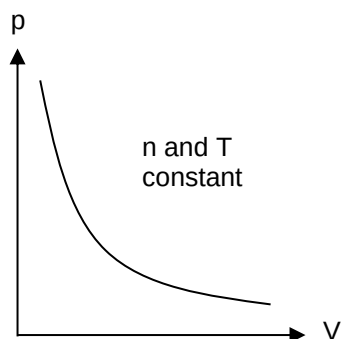
For covalent compounds, we look for cation with larger +4 charge and smaller ionic radius ($\text{Sn}^{4+} < \text{Pb}^{4+}$ in size), hence Sn^{4+} . Since Cl^- has larger electron cloud than F^- and are more easily polarised to form covalent bonds, SnCl_4 is most covalent.



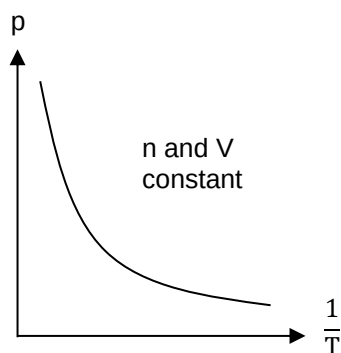
Option 2



Option 3



Option 4



16 D

$$P_1V_1 = P_2V_2$$

For O_2 ,

$$5(500) = 2(P_2)$$

$$P_2 = 1250 \text{ kPa}$$

For N_2 ,

$$3(300) = 2(P_2)$$

$$P_2 = 450 \text{ kPa}$$

By Dalton's Law, $P_T = 1250 + 450 = 1700 \text{ kPa}$

17 A

Option A (correct)

CH_4 deviates less from ideal gas behaviour as it has a smaller electron cloud size and weaker i.d.-i.d. attractions than that of CO_2 .

Option B (wrong)

NH_3 can form hydrogen bonds between molecules and there is significant intermolecular forces of attraction. NH_3 deviates more from ideal gas.

Option C (wrong)

Gas E should be a gas with a smaller molecular mass (and hence smaller number of electrons), deviating less from ideal gas.

Option D (wrong)

Gas E should form weaker i.d.-i.d. than gas F as gas E behaves more ideally than gas F.

18 A

	$2SO_2(g)$	$+ O_2(g)$	$\rightleftharpoons$	$2SO_3(g)$
Initial/mol	1	1		0
Change/mol	$-2y$	$-y$		$+2y$
Eqm/mol	$1-2y$	$1-y$		$2y$

$$\frac{2y}{1-2y+1-y+2y} = 0.50 \Rightarrow y = 0.4x$$

$$\begin{aligned} \text{Mole fraction of } SO_2 &= \frac{1-2y}{1-2y+1-y+2y} \\ &= \frac{1-2(0.4x)}{2-(0.4x)} \\ &= \frac{1}{8} \end{aligned}$$

19 A (1 and 2 only)

Option 1 (Possible)

Less OC^- ions break down. Hence equilibrium (1) lies more to the right.

Option 2 (Possible)

H^+ ions will neutralize OH^- and cause equilibrium (1) to shift forward.

Option 3 (Not possible)

Reaction (2) is not reversible.

Option 4 (Not possible)

More OH^- will cause equilibrium (1) to shift left.

20 D

% products $\downarrow$ when $P \uparrow$

i.e. eqm shifts left, since more gaseous products than reactants.

% products $\uparrow$ when $T \uparrow$,

i.e. forward rxn is endothermic.

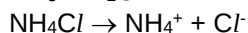
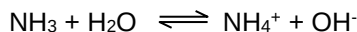
21 D

$$pH = 4.6 \Rightarrow [H^+] = 10^{-4.6}$$

$$K_a = \frac{[H^+][A^-]}{[HA]} = \frac{(10^{-4.6})^2}{0.01} = 6.31 \times 10^{-8} \text{ mol dm}^{-3}$$

22 A

An acid buffer is made up of a weak acid HA and its conjugate base A⁻. Only option 1 satisfies.

23 B

A base buffer is formed.

$$\begin{aligned} \text{pOH} &= \text{p}K_b + \log_{10} \frac{[\text{NH}_4^+]}{[\text{NH}_3]} \\ &= 4.75 + \log_{10} \frac{80 \times 0.150}{100 \times 0.100} \\ &= 4.75 + \log_{10} \frac{180}{100} \\ &= 4.75 + \log_{10} \frac{0.0667}{0.0556} \\ &= 4.83 \\ \text{pH} &= 14 - 4.83 \\ &= 9.17 \approx 9.2 \end{aligned}$$

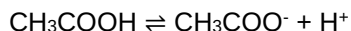
24 D

To appear red in X, pH > 5.0

To appear yellow in Y, pH < 5.7

i.e. 5.0 < pH of solution < 5.7

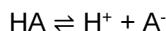
It cannot be HCl (pH = 1) or NaOH (pH = 13).



$$K_a = [\text{CH}_3\text{COO}^-][\text{H}^+] / [\text{CH}_3\text{COOH}] = x^2 / 0.10$$

$$[\text{H}^+] = x = 1.34 \times 10^{-3}$$

$$\text{pH} = -\log_{10}[\text{H}^+] = 2.87$$



$$K_a = [\text{H}^+][\text{A}^-] / [\text{HA}]$$

$$[\text{H}^+] = 5 \times 10^{-6}$$

$$\text{pH} = 5.3$$

25 C

IP > K_{sp} for precipitate to appear

$$K_{\text{sp}} \text{ of } \text{Ca}(\text{OH})_2 = [\text{Ca}^{2+}][\text{OH}^-]^2$$

$$\Rightarrow [\text{Ca}^{2+}] > (5.5 \times 10^{-6}) \div (1.0 \times 10^{-3})^2 \\ = 5.5 \text{ mol dm}^{-3}$$

$$K_{\text{sp}} \text{ of } \text{CaCO}_3 = [\text{Ca}^{2+}][\text{CO}_3^{2-}]$$

$$\Rightarrow [\text{Ca}^{2+}] > (8.7 \times 10^{-9}) \div (1.0 \times 10^{-3}) \\ = 8.7 \times 10^{-6} \text{ mol dm}^{-3}$$

$$K_{\text{sp}} \text{ of } \text{CaF}_2 = [\text{Ca}^{2+}][\text{F}^-]^2$$

$$\Rightarrow [\text{Ca}^{2+}] > (4.0 \times 10^{-11}) \div (1.0 \times 10^{-3})^2 \\ = 4.0 \times 10^{-5} \text{ mol dm}^{-3}$$

Precipitation will take place in the order of increasing concentration of Ca²⁺.

26 A

C_a is sp hybridised. The two sp hybrid orbitals are directed at 180° to each other and overlap head-on to give two σ bonds, while the two unhybridised orbitals overlap sideways to give two π bonds.

C_b is sp² hybridised. The three sp² hybrid orbitals are directed at 120° to each other and overlap head-on to give three σ bonds, while the 4th

unhybridised orbital overlaps sideways to give the π bond.

27 B**Option 1 (Wrong)**

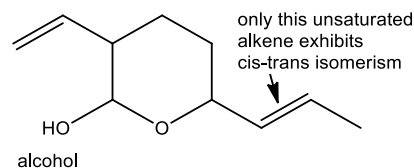
NH₃ is electron-rich (lone pair of electrons on N) and acts as a nucleophile.

Option 2 (Correct)

The nucleophile NH₃ displaces Br⁻ from CH₃CH₂Br to form the amine. Hence it follows a nucleophilic substitution mechanism.

Option 3 (Correct)

The by-product HBr reacts with NH₃ to form the salt NH₄Br via an acid-base reaction.

28 A**29 B**

UV light causes homolytic fission to produce Cl• radicals during initiation step.

Only propagation and not termination step produce HCl. To produce HCl in termination stage entails combination of H• and Cl• radicals. However, during initiation, the strong C-H bond is not broken. Hence H• radicals are not formed.

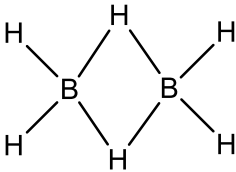
30 D

Formation of Cl• free radical causes depletion of the ozone layer. F• radicals are unlikely to be formed as the stronger C-F bond (due to the smaller size of F atom) is not broken as readily as the C-Cl bond.

	<table border="1" data-bbox="331 197 1406 360"> <tr> <td>Standard enthalpy change of combustion of $\text{CH}_3\text{C}\equiv\text{CCH}_3(\text{g})$</td><td>$-2577 \text{ kJ mol}^{-1}$</td></tr> <tr> <td>Standard enthalpy change of combustion of carbon</td><td>-394 kJ mol^{-1}</td></tr> <tr> <td>Standard enthalpy change of formation of $\text{CH}_2=\text{CHCH}=\text{CH}_2(\text{g})$</td><td>$+110 \text{ kJ mol}^{-1}$</td></tr> </table> $\text{CH}_3\text{C}\equiv\text{CCH}_3(\text{g}) + 11/2\text{O}_2(\text{g}) \rightarrow \text{CH}_2=\text{CHCH}=\text{CH}_2(\text{g}) + 11/2\text{O}_2(\text{g}) \quad \Delta H_r$ ✓✓✓✓ for energy cycle, minus 0.5 mark for every mistake By Hess' Law, ΔH_r $= \checkmark\checkmark -2577 - 4(-394) - 3(-286) + (+110)$ $= \checkmark\checkmark -33 \text{ kJ mol}^{-1}$	Standard enthalpy change of combustion of $\text{CH}_3\text{C}\equiv\text{CCH}_3(\text{g})$	$-2577 \text{ kJ mol}^{-1}$	Standard enthalpy change of combustion of carbon	-394 kJ mol^{-1}	Standard enthalpy change of formation of $\text{CH}_2=\text{CHCH}=\text{CH}_2(\text{g})$	$+110 \text{ kJ mol}^{-1}$
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	[4]						
(c)	Methanol can be manufactured cheaply and efficiently using hydrogen and carbon monoxide. The graph below shows the behaviour of 1 mol of an ideal gas at a fixed temperature. On the same axes, sketch graphs of propene, hydrogen and carbon monoxide. You may assume that the amount of each gas is 1 mol and at the same temperature as that of an ideal gas. Label each of the graphs clearly. $\checkmark\checkmark$ graph shape for real gas $\checkmark\checkmark$ H_2 shows smallest negative deviation (or very small positive deviation throughout), $\text{CH}_3\text{CH}=\text{CH}_2$ shows biggest deviation from ideality						
	[2]						
(d)	A method of measuring fuel content is to determine a quantity known as energy density, which is the heat released on burning 1 m^3 of the fuel under standard conditions. Thus the units of energy density are in J m^{-3}.						

		(i)	Calculate the energy density, in J m^{-3} for each of the organic species listed in the table above at the beginning of the question. Hence state which species has a higher fuel content. Density in g m^{-3} = density in g cm^{-3} $\times 10^6$ Amount per m^3 = density in g m^{-3} / M_r Energy density in J m^{-3} = Amount per m^3 $\times \Delta H_c^\ominus \times 1000$ For $\text{CH}_3\text{CH}=\text{CH}_2$, Energy density = $\sqrt[3]{1.74 \times 10^{-3} \times 10^6 / 42.0 \times (2058) \times 1000}$ = $8.53 \times 10^7 \text{ J m}^{-3}$ For CH_3OH, Energy density = $\sqrt[3]{0.792 \times 10^6 / 32.0 \times (715) \times 1000}$ = $1.77 \times 10^{10} \text{ J m}^{-3}$ Higher fuel content: $\sqrt[3]{\text{CH}_3\text{OH}}$
			[3]
		(ii)	With reference to the data in the table above, suggest one additional advantage of using the species you have indicated in (d)(i) as a fuel source. $\sqrt[3]{\text{CH}_3\text{OH}}$ is a liquid hence it can be transported and stored more easily.
			[1]
			[Total: 16]

S2	The following question involves borane, a highly reactive molecule with the formula BH_3 .
(a)	Draw the dot-and-cross diagram of BH_3. Use your diagram to suggest why borane is a highly reactive molecule. [1m] <u>B is electron-deficient</u> [½m] as there are only 6 electrons around it. Hence it has a high tendency to <u>form bonds to achieve a more stable octet arrangement</u>/empty orbital to accommodate electrons to achieve a more stable octet configuration. [½m]
	[2]
(b)	To achieve a more stable structure, borane forms many adducts. An example is $\text{BH}_3 \cdot \text{THF}$. The structure of THF is given below.
	THF Draw the structure of the $\text{BH}_3 \cdot \text{THF}$ adduct, stating clearly the bond angles about the central atoms involved in the formation of the adduct. [1m] <u>bond angle about B: 109.5°</u> [½m] ; <u>bond angle about O: 107°</u> [½m]
	[2]
(c)	Borane dimerises to form diborane, B_2H_6 . The synthesis of diborane in the 19 th century baffled scientists. It was not until the 1950s when its molecular structure became better known, with William Nunn Lipscomb, Jr. confirming its structure using X-ray crystallography and developing theories to explain its unique bond arrangement.

		(i)	The structure of diborane is shown below.  diborane Explain what is unique about its structure. B has three valence electrons and should only be able to form 3 single bonds but <u>each B atom forms 4 single bonds in the structure.</u> [1m] OR H has one valence electron and should only be able to form 1 single bond but <u>there are two H atoms, each shared between 2 B atoms.</u> OR Any explanation that has similar meaning to the above.	
				[1]
		(ii)	State the shape about each boron atom in diborane. The shape about each B atom is <u>tetrahedral.</u> [1m]	
				[1]
		(d)	In a closed system, diborane and borane exist in equilibrium: $\text{B}_2\text{H}_6(\text{g}) \rightleftharpoons 2\text{BH}_3(\text{g})$ The equilibrium constant value for the equation above between 353 K and 423 K is given by the following expression, where K_p is in Pa and T is in K: $\log_{10} K_p = 7.478 - \frac{6205}{T} \dots\dots(1)$	
		(i)	Write the K_p expression for the equilibrium reaction between borane and diborane. $K_p = \frac{(p(\text{BH}_3))^2}{p(\text{B}_2\text{H}_6)}$ [1m]	
				[1]
		(ii)	Using expression (1), calculate the value of K_p at 353 K and 423 K respectively. At 353 K $\log_{10} K_p = 7.478 - \frac{6205}{353} \Rightarrow K_p = 7.95 \times 10^{-11} \text{ Pa}$ [½m] At 423 K $\log_{10} K_p = 7.478 - \frac{6205}{423} \Rightarrow K_p = 6.44 \times 10^{-8} \text{ Pa}$ [½m]	
				[1]

		(iii) By considering the bonds involved in diborane and borane, state whether the forward reaction is likely to be exothermic or endothermic. Explain how the values calculated in (d)(ii) are able to confirm this. <u>The forward reaction is likely to be an endothermic reaction because bonds are being broken [1m] from diborane to borane.</u> <u>According to Le Chatelier's Principle, when temperature increases, equilibrium position shifts towards the endothermic side to counteract the increase in temperature [1m]. From the values in (d)(ii), when temperature increases, K_p increases which means that the forward reaction is favoured [1m]. This confirms that the forward reaction is an endothermic reaction.</u>	
			[3]
		The following graphs show the change in partial pressure of diborane and borane with time when a known amount of diborane is heated in a closed system. The system is said to be in <i>dynamic equilibrium</i> at t_1 min.	
		(iv) Define the term <i>dynamic equilibrium</i>. <u>A reversible reaction is said to be in dynamic equilibrium when the rate of the forward reaction equals the rate of the backward reaction [$\frac{1}{2}$m]. Hence, the concentrations/partial pressures of the reactants and products remain constant [$\frac{1}{2}$m].</u>	
			[1]
		(v) Sketch and label on the diagram above what happens to the partial pressures of diborane and borane from t_1 min to t_2 min when volume of the container suddenly increases at t_1 min at constant temperature. [Note: t_2 min refers to the time for any new equilibrium to be achieved.]	
			[2]
			[Total: 14]

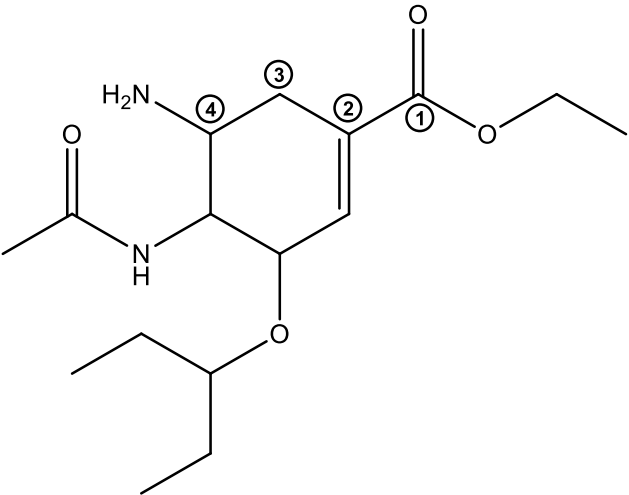
Section B: Essay Questions [40 marks]

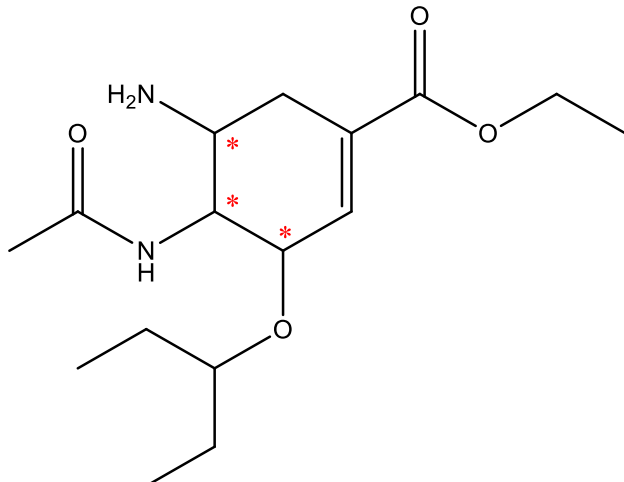
Answer **all** the questions in the spaces provided.

E1	When carbon dioxide gas dissolves in water, carbonic acid (a diprotic acid) with K_{a1} value of $4.3 \times 10^{-7} \text{ mol dm}^{-3}$ is formed. Carbonic acid is a <i>weak Bronsted-Lowry acid</i> . $\text{CO}_2(\text{g}) \rightleftharpoons \text{CO}_2(\text{aq})$ $\text{CO}_2(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{H}_2\text{CO}_3(\text{aq})$ A student was instructed to titrate 25.0 cm^3 of $0.100 \text{ mol dm}^{-3}$ carbonic acid, H_2CO_3 , with $0.0625 \text{ mol dm}^{-3}$ calcium hydroxide, $\text{Ca}(\text{OH})_2$. After the end of the titration, the student noticed that an insoluble white precipitate was formed.										
	(a)	(i)	What do you understand by the term <i>weak Bronsted-Lowry acid</i> . • Weak Bronsted-Lowry acid is a <u>proton donor</u> that <u>does not fully dissociate</u> in water.								
			[1]								
		(ii)	Calculate the pH of $0.100 \text{ mol dm}^{-3} \text{ H}_2\text{CO}_3$. Assume that the second dissociation of H_2CO_3 is negligible. • $K_{a1} = \frac{[\text{HCO}_3^-][\text{H}^+]}{[\text{H}_2\text{CO}_3]}$ $4.3 \times 10^{-7} = \frac{(10^{-\text{pH}})^2}{0.100}$ • $\therefore \text{pH} = -\lg (4.3 \times 10^{-7} \times 0.100)^{1/2} = \underline{3.68}$								
			[2]								
		(iii)	Calculate the volume of $\text{Ca}(\text{OH})_2$ needed to completely react with 25.0 cm^3 of $0.100 \text{ mol dm}^{-3} \text{ H}_2\text{CO}_3$. $\text{H}_2\text{CO}_3 + \text{Ca}(\text{OH})_2 \rightarrow \text{CaCO}_3 + 2\text{H}_2\text{O}$ $n(\text{Ca}(\text{OH})_2) = n(\text{H}_2\text{CO}_3)$ $\sqrt{0.0625 \times V} = 0.100 \times 25.0$ $\sqrt{\text{volume of Ca}(\text{OH})_2 \text{ required}} = \underline{40.0 \text{ cm}^3}$								
			[1]								
		(iv)	Calculate the pH of a solution which contains both H_2CO_3 and HCO_3^- in equal concentrations. • $K_{a1} = \frac{[\text{HCO}_3^-][\text{H}^+]}{[\text{H}_2\text{CO}_3]} \Rightarrow \text{pH} = \text{p}K_{a1} = \underline{6.37}$								
			[1]								
		(v)	The table shows some acid–base indicators. <table><tr><th>name of indicator</th><th>pH range of colour change</th></tr><tr><td>malachite green</td><td>0.2–1.8</td></tr><tr><td>bromothymol blue</td><td>6.0–7.6</td></tr><tr><td>thymolphthalein</td><td>9.3–10.6</td></tr></table>	name of indicator	pH range of colour change	malachite green	0.2–1.8	bromothymol blue	6.0–7.6	thymolphthalein	9.3–10.6
name of indicator	pH range of colour change										
malachite green	0.2–1.8										
bromothymol blue	6.0–7.6										
thymolphthalein	9.3–10.6										

		Name a suitable indicator for the above titration carried out by the student. With the aid of an equation, explain your answer. • <u>Thymolphthalein</u> • $\text{CO}_3^{2-} + \text{H}_2\text{O} \rightleftharpoons \text{HCO}_3^- + \text{OH}^-$ • <u>The working pH range of the indicator lies within the range of rapid pH change of the titration.</u> <u>[OR The pH value at the end-point coincides with the working range of the indicator.]</u>
		[3]
	(b)	Human blood is slightly basic with a normal pH of around 7.4. Its pH must be maintained for the effective transportation of oxygen. The major buffer system which controls the pH of blood is the hydrogencarbonate and carbonic acid buffer system. Explain, using equations, why an aqueous mixture of HCO_3^- and H_2CO_3 can act as a buffer in blood. • When a little H^+ is added, it is removed by a <u>large amount of HCO_3^- in the buffer. Thus pH remains almost constant.</u> $\text{HCO}_3^- + \text{H}^+ \rightarrow \text{H}_2\text{CO}_3$ • When a little OH^- is added, it is removed by a <u>large amount of H_2CO_3 in the buffer. Hence pH remains relatively constant.</u> $\text{H}_2\text{CO}_3 + \text{OH}^- \rightarrow \text{HCO}_3^- + \text{H}_2\text{O}$
		[2]
	(c)	Limewater is a saturated solution of Ca(OH)_2, and is often used to test for the presence of CO_2 formed during experiments. A positive test is indicated by the formation of a white precipitate, CaCO_3.
	(i)	Write an equation, with state symbols, to represent the reaction between limewater and CO_2. • <u>$\text{Ca(OH)}_2(\text{aq}) + \text{CO}_2(\text{g}) \rightarrow \text{CaCO}_3(\text{s}) + \text{H}_2\text{O}(\text{l})$</u>
		[1]
	(ii)	The solubility of Ca(OH)_2 is 1.73 g dm^{-3}. Calculate the concentration of Ca^{2+} in mol dm^{-3}, for a saturated Ca(OH)_2 solution. • <u>$[\text{Ca}^{2+}] \text{ of saturated solution} = \frac{1.73}{40.1+2(16.0+1.0)} = 2.33 \times 10^{-2} \text{ mol dm}^{-3}$</u>
		[1]
	(iii)	Calculate the K_{sp} of CaCO_3 given that its solubility is $1.5 \times 10^{-4} \text{ mol dm}^{-3}$. • <u>$K_{\text{sp}} = (1.5 \times 10^{-4})(1.5 \times 10^{-4}) = 2.25 \times 10^{-8} \text{ mol}^2 \text{ dm}^{-6}$</u>
		[1]
	(iv)	Hence, calculate the minimum volume of CO_2 at r.t.p., that must be bubbled through 1 cm^3 of limewater in order to give a white precipitate, CaCO_3. • <u>$2.25 \times 10^{-8} = (2.33 \times 10^{-2}) [\text{CO}_3^{2-}]$</u>

			$[\text{CO}_3^{2-}] = 9.66 \times 10^{-7} \text{ mol dm}^{-3}$ • Number of moles of $\text{CO}_2 = 9.66 \times 10^{-7} \times 1/1000 = 9.66 \times 10^{-10} \text{ mol}$ • Volume of $\text{CO}_2 = 9.66 \times 10^{-10} \times 24 = \underline{2.32 \times 10^{-8} \text{ dm}^3}$ at r.t.p.																				
			[3]																				
	(d)	When chlorine is bubbled through a solution of iodine in hot aqueous calcium hydroxide, the two halogens react in the $\text{Cl}_2 : \text{I}_2$ ratio of 7 : 1. A white precipitate, A , and a solution of calcium chloride are formed. A has the following composition by mass: Ca, 15.7%; I, 47.4%; H, 1.1%; O, 35.8%																					
	(i)	Calculate the empirical formula of A . <table><tr><td></td><td>Ca</td><td>I</td><td>H</td><td>O</td></tr><tr><td>% by mass/ g</td><td>15.7</td><td>47.4</td><td>1.1</td><td>35.8</td></tr><tr><td>Amount / mol</td><td>$15.7/40.1 = 0.392$</td><td>$47.4/126.9 = 0.374$</td><td>$1.1/1.0 = 1.1$</td><td>$35.8/16.0 = 2.24$</td></tr><tr><td>Mole ratio</td><td>1</td><td>1</td><td>3</td><td>6</td></tr></table> • Empirical formula of A is <u>CaH_3IO_6</u>			Ca	I	H	O	% by mass/ g	15.7	47.4	1.1	35.8	Amount / mol	$15.7/40.1 = 0.392$	$47.4/126.9 = 0.374$	$1.1/1.0 = 1.1$	$35.8/16.0 = 2.24$	Mole ratio	1	1	3	6
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Mole ratio	1	1	3	6																			
			[2]																				
	(ii)	Construct a balanced equation for the above reaction. • $7\text{Cl}_2(\text{g}) + \text{I}_2(\text{aq}) + 9\text{Ca}(\text{OH})_2(\text{aq}) \rightarrow 7\text{CaCl}_2(\text{aq}) + 2\text{CaH}_3\text{IO}_6(\text{s}) + 6\text{H}_2\text{O}(\text{l})$																					
			[1]																				
	(iii)	State the oxidation number of iodine in A . Let oxidation no. of iodine be x $(+2) + 3(+1) + x + 6(-2) = 0$ • Oxidation no. of iodine in A = <u>+7</u>																					
			[1]																				
			[Total: 20]																				

E2	<i>Tamiflu</i>[®] is an antiviral medication used for treatment and prevention of the flu.  <i>Tamiflu</i>[®]		
(a)	(i)	Name any four functional groups that are present in this drug. Ester, Amine, Amide, alkene (or ether). [½ mark for each correct ans. Max 2 marks]	
		[2]	
		(ii) The bond length of the carbon–carbon single bond between carbon atoms at ① and ② in <i>Tamiflu</i>[®] is 0.148 nm while the carbon–carbon single bond between carbon atoms at ③ and ④ is 0.154 nm. Explain the difference in the bond lengths. ✓The C–C single bond between carbon atoms at ① and ② is formed by overlap of <u>two sp² hybrid orbitals</u>. ✓The C–C single bond between carbon atoms at ③ and ④ is formed by overlap of <u>two sp³ hybrid orbitals</u>. ✓There is <u>greater s-character in the sp² hybrid orbital</u> than that in the sp³ hybrid orbital. ✓Thus, the electrons in the sp² hybrid orbital are <u>closer to the nucleus</u>, leading to a smaller bond length.	
		[2]	
		(iii) Does <i>Tamiflu</i>[®] display cis-trans isomerism? Explain your answer. ✓ No. This is due to ✓ring strain in the trans isomer.	
		[1]	

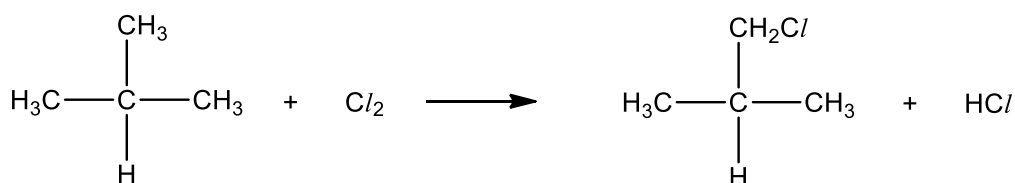
		(iv)	On the structure of <i>Tamiflu</i>[®] given below, label all the chiral centres with *. Hence determine the total number of stereoisomers of <i>Tamiflu</i>[®].  ✓ for each correct * ✓ Total number of stereoisomers = $2^3 = 8$																																
			[2]																																
	(b)	A wide range of hydrocarbons can be produced from the refining of crude oil.																																	
		(i)	10 cm³ of hydrocarbon Z, C_xH_y in the gaseous state was sparked in an oxygen-rich environment at 150 °C. There was an expansion of 10 cm³. When the gaseous product was treated with sodium hydroxide solution, there was a contraction of 30 cm³. Deduce the molecular formula of hydrocarbon Z. <table border="1" style="width: 100%;"><thead><tr><th></th><th>C_xH_y(g)</th><th>+</th><th>(x + $\frac{y}{4}$)O₂(g)</th><th>→</th><th>xCO₂(g)</th><th>+</th><th>$\frac{y}{2}$H₂O(g)</th></tr></thead><tbody><tr><td>Initial /cm³</td><td>10</td><td></td><td>10(x + $\frac{y}{4}$) + excess O₂</td><td></td><td>–</td><td></td><td>–</td></tr><tr><td>Change /cm³</td><td>–10</td><td></td><td>–10(x + $\frac{y}{4}$)</td><td></td><td>+10x</td><td></td><td>+10($\frac{y}{2}$) = +5y</td></tr><tr><td>Final /cm³</td><td>0</td><td></td><td>excess O₂</td><td></td><td>30</td><td></td><td>+5y</td></tr></tbody></table> Comparing stoichiometric coefficient, ✓10x = 30 ⇒ x = 3 Or ✓1/x = 10/30 ⇒ x = 3 Sum of change in volume = 10 ✓[–10 –10(x + $\frac{y}{4}$)] + [10x + 5y] = ✓10 [–10 –10(3 + $\frac{y}{4}$)] + [10x + 5y] = 10 		C _x H _y (g)	+	(x + $\frac{y}{4}$)O ₂ (g)	→	xCO ₂ (g)	+	$\frac{y}{2}$ H ₂ O(g)	Initial /cm ³	10		10(x + $\frac{y}{4}$) + excess O ₂		–		–	Change /cm ³	–10		–10(x + $\frac{y}{4}$)		+10x		+10($\frac{y}{2}$) = +5y	Final /cm ³	0		excess O ₂		30		+5y
	C _x H _y (g)	+	(x + $\frac{y}{4}$)O ₂ (g)	→	xCO ₂ (g)	+	$\frac{y}{2}$ H ₂ O(g)																												
Initial /cm ³	10		10(x + $\frac{y}{4}$) + excess O ₂		–		–																												
Change /cm ³	–10		–10(x + $\frac{y}{4}$)		+10x		+10($\frac{y}{2}$) = +5y																												
Final /cm ³	0		excess O ₂		30		+5y																												

		gas used if 10 cm³ of C₃H₈ was converted into a mixture of carbon dioxide and carbon monoxide in a mole ratio of 4:1. $\text{C}_3\text{H}_8 + 5\text{O}_2 \rightarrow 3\text{CO}_2 + 4\text{H}_2\text{O}$ $\text{C}_3\text{H}_8 + \frac{7}{2}\text{O}_2 \rightarrow 3\text{CO} + 4\text{H}_2\text{O}$ ✓ For 2 correct eqn For 10 cm³ C₃H₈, 8 cm³ of C₃H₈ will react to form CO₂ & H₂O while 2 cm³ of C₃H₈ will react to form CO & H₂O. Vol. of O₂ used for complete combustion ✓ = 8 × 5 (ecf) = 40 cm³ Vol. of O₂ used for incomplete combustion ✓ = 2 × 7/2 (ecf) = 7 cm³ OR = 2/2 × 7 = 7 cm³ ∴ Total volume of O₂ gas used = 40 + 7 = ✓47 cm³						
		[2]						
	(c)	Both alkanes A and B have the same molecular formula C ₆ H ₁₄ .						
	(i)	When alkane B reacted with chlorine under ultraviolet light, it produced only two isomeric monochlorinated compounds C and D, with the formula C₆H₁₃Cl. Compound D was chiral. Suggest the structures of B, C and D. <table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="width: 33%; text-align: center;">B</th> <th style="width: 33%; text-align: center;">C</th> <th style="width: 33%; text-align: center;">D</th> </tr> </thead> <tbody> <tr> <td style="text-align: center;"> ✓✓ $\begin{array}{c} \text{CH}_3 \quad \text{CH}_3 \\ \quad \\ \text{CH}_3 - \text{C} - \text{C} - \text{CH}_3 \\ \quad \\ \text{H} \quad \text{H} \end{array}$ </td> <td style="text-align: center;"> ✓ $\begin{array}{c} \text{CH}_3 \quad \text{CH}_3 \\ \quad \\ \text{CH}_3 - \text{C} - \text{C} - \text{CH}_3 \\ \quad \\ \text{H} \quad \text{Cl} \end{array}$ </td> <td style="text-align: center;"> ✓ $\begin{array}{c} \text{CH}_3 \quad \text{CH}_2\text{Cl} \\ \quad \\ \text{CH}_3 - \text{C} - \text{C}^* - \text{CH}_3 \\ \quad \\ \text{H} \quad \text{H} \end{array}$ </td> </tr> </tbody> </table>	B	C	D	✓✓ $\begin{array}{c} \text{CH}_3 \quad \text{CH}_3 \\ \quad \\ \text{CH}_3 - \text{C} - \text{C} - \text{CH}_3 \\ \quad \\ \text{H} \quad \text{H} \end{array}$	✓ $\begin{array}{c} \text{CH}_3 \quad \text{CH}_3 \\ \quad \\ \text{CH}_3 - \text{C} - \text{C} - \text{CH}_3 \\ \quad \\ \text{H} \quad \text{Cl} \end{array}$	✓ $\begin{array}{c} \text{CH}_3 \quad \text{CH}_2\text{Cl} \\ \quad \\ \text{CH}_3 - \text{C} - \text{C}^* - \text{CH}_3 \\ \quad \\ \text{H} \quad \text{H} \end{array}$
B	C	D						
✓✓ $\begin{array}{c} \text{CH}_3 \quad \text{CH}_3 \\ \quad \\ \text{CH}_3 - \text{C} - \text{C} - \text{CH}_3 \\ \quad \\ \text{H} \quad \text{H} \end{array}$	✓ $\begin{array}{c} \text{CH}_3 \quad \text{CH}_3 \\ \quad \\ \text{CH}_3 - \text{C} - \text{C} - \text{CH}_3 \\ \quad \\ \text{H} \quad \text{Cl} \end{array}$	✓ $\begin{array}{c} \text{CH}_3 \quad \text{CH}_2\text{Cl} \\ \quad \\ \text{CH}_3 - \text{C} - \text{C}^* - \text{CH}_3 \\ \quad \\ \text{H} \quad \text{H} \end{array}$						
		[2]						
	(ii)	A is a liquid that boils at 68 °C at 1 atm and it has the highest boiling point among its various structural isomers. Identify A and explain the difference in boiling points between A and B in terms of structure and bonding. ✓ A is CH₃CH₂CH₂CH₂CH₂CH₃. ✓ Both A and B have a <u>simple molecular structure</u>. ✓ From compound A to B, there is an <u>increase in branching / B is less elongated (or more spherical) in shape</u>. ✓ <u>Smaller surface area of contact between molecules B</u>. ✓ <u>Boiling point decreases from A to B as the strength of instantaneous dipole-induced dipole interaction decreases</u>. [Total 2.5, Max 2] OR						

✓ A is $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$. Having a ✓ simple molecular structure, A has the ✓ lowest degree of branching / most elongated in shape, which leads to the ✓ largest surface area of contact between molecules. Hence, A has the ✓ strongest instantaneous dipole-induced dipole interaction and the highest boiling point.
[Total 2.5, Max 2]

[2]

(d) The equation below shows how a chloroalkane can be made by the reaction of the corresponding halogen with an alkane.



Name and outline the mechanism of this reaction.

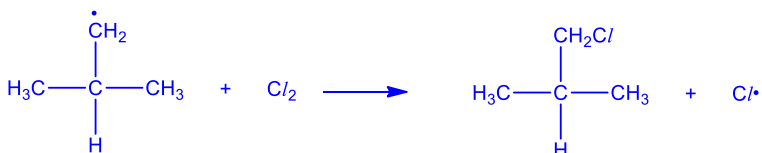
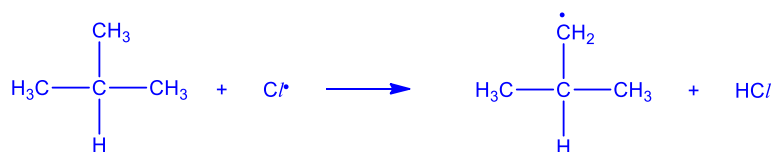
✓ Free radical substitution mechanism

Initiation:

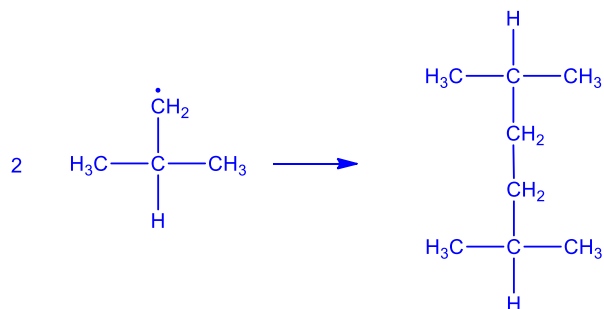
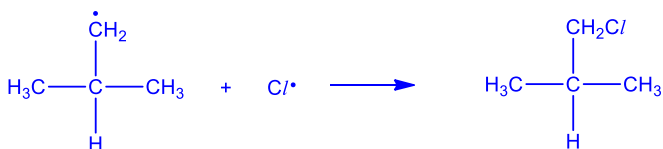


✓ Propagation:

✓✓



Termination: ✓✓ Any two



[3]

	(e) A 1,2-rearrangement is an organic reaction where a substituent, such as an alkyl group, moves from one atom to another atom in a chemical compound. In a 1,2 shift, the movement involves two adjacent atoms but moves over larger distances are possible. Below shows an example of the 1,2-rearrangement of the carbocation formed after an alkane undergoes heterolytic fission. carbocation P carbocation Q By referring to the relative stability of the carbocations P and Q, explain why 1,2-rearrangement is a favourable mechanism. ✓<u>P – 2° carbocation and Q – 3° carbocation</u> ✓<u>Greater number of alkyl groups attached to C with positive charge, hence</u> ✓<u>stronger electron-donating inductive effect exerted, ✓positive charge is dispersed to a greater extent and ✓stabilises carbocation.</u> Thus mechanism (1,2-rearrangement of an alkyl group) is favoured due to the formation of <u>more stable carbocation.</u> <u>[Total 2.5, Max 2]</u>
	[2]
	[Total: 20]